MDM-230-COM: Digital Electronics and Logic Design

Unit I – Boolean Algebraand Simplification Techniques

Boolean **Algebra:** Basic theorems and properties of Boolean algebra, De Morgan's rules. **Signed Binary number representation:** Signed Magnitude, 1's complement and 2's complement Representation.

Simplification Techniques: Sum of product and Product of sum form, Minimization of SOPandPOS using K-map. Minimization of Boolean function using K-map (up to 4 variables).

1) Simplify following logic function using Boolean algebra

$$f = A\bar{B}C + B\bar{C}D + \bar{A}BC$$

2) A staircase light is controlled by two switches, one at the top of the stairs and another at the bottom of stairs.

a) Make a truth table for this system
b) Write the logic equation in SOP form
c) Realize the circuit using AND-OR gates
d) Realize the circuit using NAND gates only

3) Represent +27, - 62 in

1) Signed magnitude
2) 1's complement
3) 2's complement

4) Represent +87, - 43 in

1) Signed magnitude
2) 1's complement
3) 2's complement

5) Convert following expression to canonical SOP form and minimize it using K-Map Y = AB + AC + BCD

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6) Convert following expression to canonical POS form and minimize it using K-Map Y = (A+C+D) (B+C+D)(B+D)

7) Minimize the following function in POS form using K-

Map: F1 (A, B, C, D) = $\prod$ M (0, 1, 2, 3, 11, 12, 14, 15)

8) Minimize the following function in SOP form using K-Map:

F1 (A, B, C, D) = $\sum$ m (0, 4, 5, 10, 11, 13, 15)+d(6, 14)

9) Minimize the following function in POS form using K-Map:

F1 (A, B, C, D) = $\prod$ M (1, 2, 8, 9, 10, 11, 14) . d (7, 15)

10) Minimize the following function in SOP form using K-
Map: 11) f (A, B, C, D) = $\sum$ m (0, 4, 5, 10, 11, 13, 15)

Unit II - Combinational Logic Design

Adders / Sub tractors: Half Adder, Full Adder, Half Subtractor, Full Subtractor, Binary Adder (IC7483), BCD adder, Look ahead carry generator.

Code Converters: BCD, Excess-3, Gray code, Binary Code. Multiplexers and Demultiplexers, Comparators (2 bit).

1) Design Full-adder circuit using two Half-adders. In diagram use block diagramof half-adder.

2) Design Full-subtractor circuit using two Half-sub tractors. In diagram use block diagramofhalf-subtractors.

3) Design single digit BCD adder using 4-bit binary adder IC7483 and explain two examplesofBCDaddition.

4) Write a short note on look ahead carry generator also explain its importance. 5) Design BCD toExcess-3code converter using minimum number of gates.

6) Design BCD to Excess-3 code converter using minimum Number of NAND gates.

7) Design Excess-3 to BCD code converter using minimum number of gates.

8) Design Excess-3 to BCD code converter using minimum Number of NAND gates.

9) Design 4-bit Binary to Gray code converter using minimum number of gates. 10) Design 4-bit Gray to Binary code converter using minimum number of gates.

Unit-III: Sequential Logic Design

Flip-Flops: SR, JK, D, T; Preset &Clear, Master Slave JK Flip Flops, Truth Tables and Excitationtables,Conversion from one type to another type of Flop Flop.

Registers: Registers: SISO, SIPO, PISO, PIPO, Shift Registers

Counters: Ring Counter, BCD Counter, Johnson Counter.

1) Distinguish between combinational and sequential switching circuits also write examples of both.

2) Convert Following Flip flops

i) SR to T ii) JK to D

3) What do you mean by excitation table of flip flop? Write the excitation table of i) S-R flips flop ii) J-K flips flop

4) With neat diagrams explain the working of the following types of shift registers i) Serial-in, serial-out ii) Parallel-in, serial-out

5) What are the advantages of M S JK flip flop? Explain the working of MS J-K flip flop in detail.

6) State different types of shift registers. Give its applications.

7) Compare synchronous and Asynchronous counter.

8) Explain how JK flip is converted into.

i) D FF ii) T FF

9) Write short note on Ring counter

10) What are the applications of Flip-Flop? Explain shift register

11) Explain S-R Flip-Flop and J-K Flip Flop using NOR gate.

12) Convert the following Flip Flops . i) D to T ii) JK to D_4

Unit IV - Algorithmic State Machines and Logic Families

Algorithmic State Machines: Finite State Machines (FSM) and ASM, ASMcharts, notations, constructionofASM chart and realization for sequential circuits.

Logic Families: Classification of logic families: Unipolar and Bipolar Logic Families, CharacteristicsofDigital ICs: Fan-in, Fan-out, Current and voltage parameters, Noise immunity, PropagationDelay, PowerDissipation, Figure of Merits, Operating Temperature Range, power supply requirements, IntroductiontoTTL& CMOS, Comparison between TTL and CMOS

1) Draw the state diagram, state table, and ASM chart for a 2-bit binary counter havingoneenable line E such that E = l counting enabled, and E = 0 counting disabled.

2) What is an ASM Chart? Name the elements of an ASM chart and define each of them.

3) Draw the ASM chart for a 2-bit UP-DOWN counter with mode select line Msuchthat: M=0 (UP Counting), M = 1 (DOWN Counting).

4) What is an ASM Chart? Name the elements of an ASM chart and define each of them.

5) What is ASM chart? Design ASM chart for 3-Bit gray code sequence withup-downconditions.

6) What is ASM Chart? State & Explain basic components of ASMchart. Alsoexplainthesalient features of ASM chart.

7) Draw and explain various notations used in ASM charts

8) Compare between FSM and ASM

9) With the help of a neat diagram, explain the working of two-input TTL NANDgate.

10) What is the advantage of open collector output? Justify your answer with suitable circuit.

11) Compare TTL and CMOS logic family.

12) What is logic family? Give the classification of logic family and also writeimportantcharacteristics of CMOS.

13) Explain TTL open collector.

14) Define the following terms and mention the standard values for TTL logic Familyi) Fan-out

ii) Power Dissipation

iii) Propagation Delay.

iv) Noise margin

v) Figure of Merit

Unit V - Programmable Logic Devices

PLDS: PLD, ROM as PLD, Programmable Logic Array (PLA): Implementation procedure, Constructionand working, Advantages over read only memory and applications, Programmable Array Logic (PAL): Architecture, Programming PAL's, construction and working, Designing combinational circuits usingPLDs.

1) Implement following Boolean function using PAL

$$F1 = \Sigma m\ (0, 2, 4, 6, 8, 12)$$

$$F2 = \Sigma m\ (2, 3, 8, 9, 12, 13)$$

$$F3 = \Sigma m\ (l, 3, 4, 6, 9, 11, 12, 14, 15)$$

2) Draw a block diagram of the PLA device and explain.

3) Implement BCD to Excess-3 code converter using PAL.

4) What is the difference between PAL and PLA?

5) Draw and explain the general structure of PLA.

6) Implement following Boolean function using PAL

$$F\ (A, B, C, D) = \Sigma\ m\ (0, 1, 3, 15)$$

7) Implement BCD to Ex-3 code converter using PAL.

8) What is the difference between PAL and PLA?

9) Implement the following Boolean function using PAL.

$$F1\ \Sigma\ m\ (0, 3, 5, 7, 9, 10, 11, 14, 15)$$

F2 Σ m (2, 3, 12, 14)

10) Draw block diagram of PLA device and explain in detail.

11) Implement gray to binary code using PLA.

12) Implement the following using PROM

$$A(X, Y, Z) = \sum m (5, 6, 7)$$

$$B(X, Y, Z) = \sum m (3, 5, 6, 7)$$

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13) Implement the following Boolean functions using PAL

X1(A, B, C, D) Σ m (0, 2, 6, 7, 8, 9, 12, 13)

X2(A, B, C, D) Σ m (3, 6, 7, 11, 14, 15)